

ASPECTS OF COLOUR MANAGEMENT

Part 1 - CHARACTERISATION OF THREE COLOUR IMAGING DEVICES

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1. INTRODUCTION

Problems in predicting and controlling colour have become increasingly prevalent in recent years across a variety of industries. In the past, many of these industries have relied on empirical techniques utilising the skills of highly trained staff. However, this is not an efficient or predictable procedure and the development of tools such as colorant formulation systems in the decorating industries, or standardisation in the reproduction industries, have been directed at improving the efficiency and predictability of the products. These techniques rely heavily on colorimetry and have proved very successful in achieving their objectives. Unfortunately, they are only fully effective in somewhat limited circumstances.

Where absolute matching and control of colour is required these techniques prove very efficacious. This has been the primary area of application for them in most of the colorant industries in the past. However, where matching across quite different media is required, in which the colour gamuts of the media may be quite different, the selection or interpretation of colour has still required the skills referred to above and for critical colour matching has relied on these staff physically producing the colours in such a way that the results closely simulate those which will finally be rendered so that a proper subjective assessment may be made. In the relatively rare situations where such a simulation (or proof) was not possible the user had to mentally interpret the result and this could again lead to somewhat unpredictable results depending on the skills and experience of the user.

In recent years, a number of events have come together to make the problems associated with this interpretation or simulation more acute. The 'explosion' of colour in everyday life has led to an insufficient number of such skilled staff and to more critical consumers who expect to see the same colour appearance across a whole range of related products. Additionally, the development of Computer Aided Design (CAD) systems for use in many colour industries has led to serious difficulties where users, who may lack the interpretive skills, expect to see the colour on the screen reproduced accurately on the various media they are designing for. Furthermore, the improvements in communication systems are leading to working situations where colours, in the form of images or dye recipes, are being transmitted to a remote site with no hard copy to act as a reference. This places an increasing reliance on the 'numbers' to properly predict the appearance of the colour. Unfortunately, the prediction of appearance is where colorimetry, based on the CIE system¹, is not so effective.

Two consecutive projects, MMI/146 and IED4/1/1201 (CARISMA) funded by the British government's Alvey and IEATP programmes respectively, were carried out to overcome these problems during the period 1986-1992. The consortium included Coats Viyella plc, Crosfield Electronics Ltd., Sigmex Displays, and Loughborough University representing a range of industries and interests who each recognised the limitations of the current techniques. The primary aim of these projects was to increase knowledge in the domains of colour appearance modelling, various aspects of colour reproduction and colouring technology and the communication of colour. Four main tasks were specified:

- 1) to derive a reliable of colour vision which can accurately predict the appearance of colours presented on multiple media under different lighting conditions;
- 2) to define and implement a procedure for establishing the mapping between computer-controlled imaging devices which best approximates equality of colour appearance;
- 3) to research and implement methods for formulation of non-process inks and dyes that match the target colours displayed on a monitor;
- 4) to provide a graphical user interface at a workstation that allows users to visually select and navigate the principal colour order systems and colour science based models.

The work described here summarises the second task, which was concerned with two specific problems: *device characterisation* and *gamut compression*. These issues are of fundamental importance in defining a colour management system which uses colorimetry as the basis for controlling the colour match between an original and its reproduction. Device characterisation is a term which has crept into colour reproduction parlance in recent years and has slightly different meanings to different people. For the purposes of this project it has been defined as the provision of data to enable development of a mathematical transformation which determines the conversion between device specific data and colorimetric data based on the CIE system¹. For output devices the resultant transformation provides a technique for defining the 'amounts' of colorant, or colour monitor drive voltages, required to produce a colour with specific tristimulus values. For input devices it provides a method for obtaining tristimulus values from the signals generated when scanning an image or creating it on a colour monitor.

However, a large number of visual results obtained from task one (LUTCHI Colour Appearance Data)²⁻⁶ show that simply generating tristimulus values of an original image and reproducing those on output will not always produce a colour match. If the media are very different (such as photographic film, colour monitors and reflection prints) the viewing conditions and texture of the media influence the appearance of the colour. The tristimulus values have to be 'corrected' to compensate for this. Hence, it was concluded that a 'four stage transform'⁷ which takes account of these effects is required to achieve a colour match between two colours on dissimilar media. This is summarised in figure 1 and explained below:

- Stage 1 Transform 'raw' image data from a scanner or CAD system into tristimulus values for a specific illuminant and observer (e.g. D50 and 2 degree).
- Stage 2 Convert tristimulus values into hue, lightness and chroma using a colour appearance model with appropriate parameters for the viewing condition used for assessing the original image.
- Stage 3 Convert hue, lightness and chroma into tristimulus values using the colour appearance model with appropriate parameters for the viewing condition used for assessing the rendered image and corrections for media textural differences.
- Stage 4 Transform tristimulus values into the signals required to generate the colour on the output device. These will be ink amounts for printing, exposure levels for transparency recording or drive voltages for a colour monitor.

For stages two and three, the Hunt colour appearance model ^{8,9} is recommended. (This model was developed in task one and correlates well with the LUTCHI Colour Appearance Data²⁻⁶.) Thus, stages two and three of the four stage transform are not covered in this paper which is concerned with stages one and four.

Two approaches were been taken to output device characterisation in this work. Initially, a *full characterisation* method was developed where a large number of colours were measured to define the transformation. This gave us accurate transformations for a range of device types as described in section two. However, such a procedure is time consuming and so we have looked to develop *modelling* methods which require far less measurement as discussed in section three. Generally they are less accurate but are often adequate for the process required. Where they are not they can still be used in a differential mode to correct the full characterisation when the device characteristics have changed slightly. Section 2 summarises the devices selected and discusses the full characterisation. In section 3 the models used for characterisation of the three colour devices are described.

This paper is divided into two parts. Part 1 is concerned with the principles of device characterisation and discusses the results obtained when characterising three colour devices. In part 2 this is extended to four colour devices and combined with a discussion of a gamut compression study; an essential element of any colour reproduction scheme.

2. DEVICE CHARACTERISATION

The purpose of device characterisation is to provide the data to define a mathematical transform which enables the conversion between CIE data and the device signals. We consider two categories of characterisation:

- a) The first, associated with input devices, enables the user to generate a digital file, which is encoded in such a way that the colour is unambiguously specified, either by scanning the image on some type of 'digitiser' or by 'creation' on a colour monitor, followed by appropriate data processing.
- b) The second, associated with output devices, enables the user to render an image, according to the tristimulus values of an image file and after correction for the appropriate parameters for appearance, onto hard copy (e.g. paper or film) or a colour monitor.

2.1 DEVICES SELECTED

At the outset of the project a number of devices were selected as being 'typical' of those which may be encountered across a whole gamut of cost and quality. Two input scanners (or digitisers) were selected: one representing photomultiplier and external drum technology manufactured by Crosfield and another, manufactured by Sharp, representing the more common 'desktop' scanners using CCD and flatbed technology. Six hard copy output devices were selected. DuPont Cromalin proofing of conventional halftone colour separations represented the typical Graphic Arts quality, with two ink-jet technologies (from Iris and Hewlett Packard) representing the high and low cost ink jet products. Thermal printers were represented by the Mitsubishi G650, Electrophotographic printers by the Canon CLC500 and film transparency recorders by the CSI Fire 1000. The Barco Calibrator was selected to represent colour monitors. This is an expensive, high quality monitor but despite that we felt it was only necessary to characterise one since monitor technology and characteristics do not differ as greatly as for input scanners or output printers. Thus the procedure developed for the Barco should be applicable to other monitors albeit that they may have a rather different gamma and matrix transformation and be more variable from day to day.

2.2 CHARACTERISATION PROCEDURE

2.2.1 Hard copy devices

The procedure for full characterisation of paper hard copy devices, regardless of the technology employed, was to initially consider only three primaries (cyan, magenta, and yellow). The inclusion of the black colorant is considered to be a simple extension of this and will be covered in part 2 of this paper. Essentially, the characterisation procedure employed in this work can be defined as a series of steps as given below:

- a) Produce a series of steps of increasing density for each colorant used by the device. Unless one has prior knowledge there would normally be a reasonably large number of these steps produced with equal quantisation intervals.
- b) Measure the samples and calculate the lightness of each step. Plot lightness as a function of the quantisation level. (To simplify the measurement procedure the use of density, analytical or colorimetric, rather than lightness is generally a good enough approximation). If the relationship between quantisation level and lightness (or density) is highly non-linear this is used to define a single dimensional look up table which will be added to the three dimensional look up table described in step e).
- c) Select nine quantisation levels such that the lowest and highest values are included and the seven intermediate values provide eight approximately equal intervals in lightness or density. (This ensures that the colour data used for characterisation gives a reasonably uniform perceived colour difference between adjacent levels).
- d) Print a series of test images which includes each combination of the nine levels for each primary colorant for colorimetric measurement. This is normally achieved by printing squares with approximately 1" sides since these are large enough for measurement on spectrophotometers with reasonably large measuring apertures to avoid errors arising from internal scattering in the substrate and local print non-uniformity. Nine prints are produced as shown in Fig. 2; each with a different level of one of the colorants. The full range of the other two colorants are overprinted on each of these levels such that nine rows of one overprint nine columns of the other. In this way 729 colours are produced. Ideally a number of replicate prints are made to ensure the samples are representative and measurements averaged.
- e) Each colour is measured and the resultant data is used to generate a 'look-up table' (LUT) from which the quantisation levels required to drive the device can be 'read out' in order to achieve a specific colour from the 729 measured. For intermediate colours a linear interpolation may be used. (This was why the data was 'linearised' in steps a) and b)).

This data is denoted the *cube data set* representing the device characterisation information. Basically, it consists of 512 sub-cubes, i.e. 8^3 . The mathematical transform for the CIE tristimulus values from a given set of CMY primaries is obtained by finding the sub-cube in which the sample point falls and then, from the tristimulus values of the corner points of the sub-cube, using the Stone et al¹⁰ trilinear interpolation method given in equation 1.

$$\begin{aligned}
 X &= (1-D_c)(1-D_m)(1-D_y) X_1 + D_c(1-D_m)(1-D_y) X_2 + (1-D_c) D_m(1-D_y) X_3 + D_c D_m(1-D_y) X_4 + \\
 &\quad (1-D_c)(1-D_m) D_y X_5 + D_c(1-D_m) D_y X_6 + (1-D_c) D_m D_y X_7 + D_c D_m D_y X_8 \\
 Y &= (1-D_c)(1-D_m)(1-D_y) Y_1 + D_c(1-D_m)(1-D_y) Y_2 + (1-D_c) D_m(1-D_y) Y_3 + D_c D_m(1-D_y) Y_4 + \\
 &\quad (1-D_c)(1-D_m) D_y Y_5 + D_c(1-D_m) D_y Y_6 + (1-D_c) D_m D_y Y_7 + D_c D_m D_y Y_8 \\
 Z &= (1-D_c)(1-D_m)(1-D_y) Z_1 + D_c(1-D_m)(1-D_y) Z_2 + (1-D_c) D_m(1-D_y) Z_3 + D_c D_m(1-D_y) Z_4 + \\
 &\quad (1-D_c)(1-D_m) D_y Z_5 + D_c(1-D_m) D_y Z_6 + (1-D_c) D_m D_y Z_7 + D_c D_m D_y Z_8
 \end{aligned} \tag{1}$$

where D_c , D_m , and D_y are the distances from the origin of the selected sub-cube to the input CMY primaries. X_i , Y_i and Z_i are the tristimulus values for the eight points in this particular sub-cube.

In a similar way the CMY values needed to produce a specific set of tristimulus values may be obtained. To achieve this CMY replaces XYZ in the equations above.

This procedure is the most accurate transformation technique (for this level of measurement) if the data is reasonably well linearised and the test prints are truly representative of the device's output. If it is necessary to increase the accuracy further the number of steps of each colorant used to generate the test

images may be increased, with the consequent increase in measurement. However, such a necessity normally implies inadequate linearisation of the data and a preferable approach is to correct this. Alternatively non-linear interpolation may be employed.

2.2.2 Input scanners and transparency recorders

For these devices, only a modelling method was used because full characterisation is not very practical using LUT and interpolation techniques. For reflection copy materials size is something of an issue and for transparency materials measurement is more difficult than for paper hard copy. We thus chose a smaller set of data and modelled these devices. Discussion of these is therefore deferred to section three.

2.2.3 Colour monitors

The method for characterising the monitors was initially the same as that of hard copy devices. However, it was found that the accuracy of the technique was rather poor and although a full characterisation was generated using the 729 colours obtained by selecting all combinations of nine levels of each primary the results were not as good as those obtained by modelling as will be discussed in section 3.3. This was due to inadequate linearisation of the device. Since we found that adequate results could be achieved by modelling, providing sufficient points were used for linearisation, a full characterisation using proper linearisation and perceptually uniform steps was not attempted..

2.3 Measuring instrument correlation

Whilst all of the studies to obtain the LUTCHI Colour Appearance Data had been carried out using a telespectroradiometer (TSR) it was convenient for the reflection hard copy devices to use conventional spectrophotometry. Since part of the project involved development of an exemplar, showing a practical colour management system in operation, it was necessary to reconcile these two sets of measurements. To this end a range of reflection print samples were produced and measured using both instruments. Methods for correlating the data from these instruments were developed. It was found that a second-order 'best fit' polynomial, specific to each material, provided an acceptable transformation between the different instrumental reflectance values. This method was used to convert all the colorimetric data measured using the spectrophotometer to those of the TSR when using the four stage transform, described in the introduction, to view and compare images on different media.

3. DEVICE MODELLING

In general, the amount of measurement implied by the procedure described in section 2.2.1 is impractical; particularly in an industrial environment. Where processes are perfectly stable, or a separate and reliable calibration procedure which returns a process to its original state exists, it may be justifiable, but this is rarely the case. Characterisations therefore have to be computed fairly regularly and so we decided that it was desirable to investigate some simple modelling techniques which require less measurement. The models investigated are described in section 3.1, the results obtained from them are discussed in section 3.2.

3.1 MODELS

Each model derived in this project was required in two forms. *Forward* models were derived for predicting the CIE tristimulus values from a set of device primaries (i.e. RGB for monitors, scanners, and film output recorders, and CMY for printing devices). The *reverse* models were also derived to

obtain the device primaries from a corresponding set of tristimulus values. Only the forward model is briefly described here; the principle is precisely the same for derivation of the reverse models.

3.1.1 Monitor Models

The model in common use in the Graphics Arts industry is to assume a linear gamma function and employ a matrix transformation to convert tristimulus values to drive voltage. However, we know this is not adequate for the highest quality levels. We, therefore decided to investigate a number of alternative procedures. This study closely followed the work of Post and Calhoun¹¹.

The typical relationship between the monitor luminance in each of the R, G, and B channels and the CIE XYZ tristimulus values is given by the matrix expression:

$$\begin{matrix} X \\ Y \\ Z \end{matrix} = \begin{matrix} x_r/y_r & x_g/y_g & x_b/y_b \\ 1 & 1 & 1 \\ z_r/y_r & z_g/y_g & z_b/y_b \end{matrix} \begin{matrix} T_r \\ T_g \\ T_b \end{matrix} \quad (2)$$

where T_r , T_g and T_b are the luminances and x_r , y_r , z_r ; x_g , y_g , z_g and x_b , y_b , z_b are the chromaticity co-ordinates respectively for each of the red, green and blue channels. Six models were derived the first five of which assume that the chromaticity coordinates for each channel are invariant and use the matrix above. Each of these five models simply transforms the normalised DAC value (D) to the normalised luminance for each of the three channels (T in equation 2). Both normalised scales are ranged from zero to one assuming zero luminance at zero DAC value. A brief account of each model is given below:

1) PLCC (Piecewise Linear interpolation assuming constant Chromaticity Co-ordinates) This model defines a look up table specifying the luminance values measured for given DAC values, combined with linear interpolation for intermediate values. It is assumed that the luminance in each channel changes linearly between the points defined in the table.

2) LIN-LIN2 (linear-linear second-order model)

This model assumes that the relationship between D and T for each channel is given by equation 3.

$$T = c_1 + c_2 D + c_3 D^2 \quad (3)$$

where c_i are the optimised coefficients for a particular channel obtained by least squares fitting from the measured data.

3) LOG-LOG (log-log model)

This model, equation 4, has been widely used in the display industry and is more generally known as gamma correction.

$$\log T = c \log D \quad (4)$$

where c is the optimised coefficient for a particular channel obtained by least squares fitting from the measured data.

4) LOG-LOG2 (log-log second-order model)

This model was devised by Cowan¹². It assumes that the relationship between D and T for each channel is as given in equation 5 and is an attempt to correct for the departures from linearity, particularly at low luminance levels, assumed in the gamma correction model described above.

$$\log T = c_1 + c_2 \log D + c_3 (\log D)^2 \quad (5)$$

where c_i are the optimised coefficients for a particular channel obtained by least squares fitting from the measured data.

5) LOG-LIN2 (log-linear second-order model)

This model assumes that the relationship between D and T for each channel is given by equation 6.

$$\log T = c_1 + c_2 D + c_3 D^2 \quad (6)$$

where c_i are the optimised coefficients for a particular channel obtained by least squares fitting from the measured data.

6) PLVC (Piecwise Linear interpolation assuming Variable Chromaticity co-ordinates) This model is similar to PLCC except the chromaticity coordinates in each channel are assumed to change as a function of luminance. A 729 colour cube data set combined with trilinear interpolation, as described in section 2.2.1 was used. This model was expected to give the most accurate correlation between the DAC values and colorimetric results, because it takes into account the problems of gun interdependence and phosphor constancy. However, it requires much more complex algorithms and calibration procedures than the other models.

For the regression models, two to five, the nine DAC levels for each individual channel and the corresponding luminance values from the characterisation set of each monitor were used to optimise the coefficients in each model using the Quasi-Newton optimisation technique¹³.

3.1.2 Printer Models

Two types of model were used for these devices. Both are well described in the literature and some of the limitations are well known. However, previous studies by one of the authors¹⁴ had shown that they can work well in certain circumstances and it was decided that their applicability to the devices being used in this study should be investigated.

Neugebauer equations¹⁵ are appropriate for halftone process and assume that the halftone dots are randomly distributed on the print. Whilst not strictly correct the screen angles and rulings commonly used make this a reasonable approximation. Masking equations¹⁶ are appropriate to any device in which the density of the colorants are additive. This is generally the case for non-turbid, but not for turbid, media. Certain corrections can be made for these errors as will be described.

a) Neugebauer equations

The Neugebauer equations are based on the assumption that the colour of a unit area is given by the additive mixture of the tristimulus values of the substrate and each individual primary colour and overprint arising on a micro scale within the print. It is assumed that the ink film thickness is uniform so that these values may be measured on large areas of the appropriate ink combinations and correlated with those occurring at the micro level. For any particular combination of inks it is assumed that there is an equal probability of any specific primary or overprint occurring which is directly related to the sizes of the individual halftone dots.

From three colours there are eight possible combinations, the three primary colours (cyan, magenta, yellow), overprints of each pair of colours (red, green, blue), the white paper where no colour is printing (white) and the overprint of the three colours (black). Hence, only eight samples are needed to derive these equations. Thus it is possible to compute the tristimulus value of the reproduction by adding together the tristimulus values of each of the eight components of the half-tone structure which make up that region multiplied by the appropriate fractional dot areas. The formula is given in equation 7.

$$X = (1-c)(1-m)(1-y)X_w + c(1-m)(1-y)X_c + (1-c)m(1-y)X_m + (1-c)(1-m)yX_y + c m (1-y)X_b + c(1-m)yX_g + (1-c)m y X_r + c m y X_{black}$$

$$Y = (1-c)(1-m)(1-y)Y_w + c(1-m)(1-y)Y_c + (1-c)m(1-y)Y_m + (1-c)(1-m)yY_y + c m(1-y)Y_b + c(1-m)yY_g + (1-c)m y Y_r + c m y Y_{black}$$

$$Z = (1-c)(1-m)(1-y)Z_w + c(1-m)(1-y)Z_c + (1-c)m(1-y)Z_m + (1-c)(1-m)yZ_y + c m(1-y)Z_b + c(1-m)yZ_g + (1-c)m y Z_r + c m y Z_{black} \quad (7)$$

where X, Y, Z are the tristimulus values of the colour to be matched and X_w, X_c etc. are the X tristimulus values of the eight component colours. The c, m and y values are the fractional dot areas on paper of the three inks which may be calculated using the Murray-Davies equation¹⁷ given in equation 8.

$$\begin{aligned} c &= (1 - 10^{-D_c}) / (1 - 10^{-D_{cs}}) \\ m &= (1 - 10^{-D_m}) / (1 - 10^{-D_{ms}}) \\ y &= (1 - 10^{-D_y}) / (1 - 10^{-D_{ys}}) \end{aligned} \quad (8)$$

where D_c = log(X_o/X), D_{cs} = log(X_o/X_s), D_m = log(Y_o/Y), D_{ms} = log(Y_o/Y_s), D_y = log(Z_o/Z), D_{ys} = log(Z_o/Z_s) and (X_o, Y_o, Z_o), (X_s, Y_s, Z_s) and (X, Y, Z) are the tristimulus values for the substrate, solid ink, and sample respectively.

Unfortunately, there are a number of problems with this model. There seems little doubt that the major one is that which arises from light scatter within the substrate which means that the Murray-Davies equation does not give an accurate prediction of the geometric area of the halftone dots. However, this light scatter, which has the effect of increasing the density of the print, is part of the process of light interaction with the media which affects visual perception and must therefore be accounted for. In this sense the Murray-Davies equation is the appropriate one to use but it does not help in understanding the area of dot required on the colour separation in order to achieve the required colour since the effects of light scatter and physical changes in the dot size during imaging and printing cannot be separated.

In many ways the most practical solution to this problem is to print a scale of halftone steps in each of the primary colours and measure the dot area on the print using the Murray-Davies equation. This can then be used to define a look-up table which converts Murray-Davies dot area to film dot area and takes account of both light scattering and physical changes. Previous studies¹⁴ have shown that such a technique gives quite accurate results. However, a more common approach is to use the Yule-Nielsen modification to the Murray-Davies equation¹⁸ or the Yule-Colt modification to the Neugebauer equations¹⁹ both of which introduce a factor n, which is derived empirically, to correct for these errors.

The Yule and Colt model includes an exponent (1/n) for the tristimulus values in both sides of equation 7. The n value varies according to the different printing devices' primaries, paper, screen frequency, and dot area used. This model is given in equation 9.

$$\begin{aligned} X^{1/nc} &= (1-c)(1-m)(1-y)X_w^{1/nc} + c(1-m)(1-y)X_c^{1/nc} + (1-c)m(1-y)X_m^{1/nc} + \\ &\quad (1-c)(1-m)yX_y^{1/nc} + c m(1-y)X_b^{1/nc} + c(1-m)yX_g^{1/nc} + (1-c)m y X_r^{1/nc} + \\ &\quad c m y X_{black}^{1/nc} \\ Y^{1/nm} &= (1-c)(1-m)(1-y)Y_w^{1/nm} + c(1-m)(1-y)Y_c^{1/nm} + (1-c)m(1-y)Y_m^{1/nm} + \\ &\quad (1-c)(1-m)yY_y^{1/nm} + c m(1-y)Y_b^{1/nm} + c(1-m)yY_g^{1/nm} + (1-c)m y Y_r^{1/nm} + \\ &\quad c m y Y_{black}^{1/nm} \\ Z^{1/ny} &= (1-c)(1-m)(1-y)Z_w^{1/ny} + c(1-m)(1-y)Z_c^{1/ny} + (1-c)m(1-y)Z_m^{1/ny} + \\ &\quad (1-c)(1-m)yZ_y^{1/ny} + c m(1-y)Z_b^{1/ny} + c(1-m)yZ_g^{1/ny} + (1-c)m y Z_r^{1/ny} + \\ &\quad c m y Z_{black}^{1/ny} \end{aligned} \quad (9)$$

In this study n_c , n_m and n_y were obtained by employing an empirical optimisation technique using the cube data set until the minimum mean $CMC(1:1)\Delta E$ was obtained between the predicted and measured results. This model is denoted the modified Neugebauer model.

b) Masking equations

These equations were suggested by Yule¹⁶ and assume additivity and proportionality of ink densities as the halftone dot area, or colorant concentration or film thickness changes. If this were correct a simple linear transformation would be sufficient to establish the amount of ink required in the reproduction in order to match the three colour intensities in the original copy. However, we know that serious departures from additivity and proportionality arise in practice particularly because of the half-tone screen and turbidity of the media. Hence, Clapper²⁰ suggested that the masking equations can be expanded to include second-order and cross-product terms in order to give a more accurate prediction than the simple first-order equations. Yule²¹ subsequently suggested that greater accuracy could be obtained by even higher order polynomials such as third-order. The third-order form of the masking equations are given in equation 10. The second order form consists of the first 9 terms only.

$$\begin{aligned}
 D_r &= a_1 c + a_2 m + a_3 y + a_4 c^2 + a_5 m^2 + a_6 y^2 + a_7 c m + a_8 c y + a_9 m y + a_{10} c^3 + \\
 &\quad a_{11} m^3 + a_{12} y^3 + a_{13} c^2 m + a_{14} c^2 y + a_{15} m^2 c + a_{16} m^2 y + a_{17} y^2 c + a_{18} y^2 m + \\
 &\quad a_{19} c m y \\
 D_g &= b_1 c + b_2 m + b_3 y + b_4 c^2 + b_5 m^2 + b_6 y^2 + b_7 c m + b_8 c y + b_9 m y + b_{10} c^3 + \\
 &\quad b_{11} m^3 + b_{12} y^3 + b_{13} c^2 m + b_{14} c^2 y + b_{15} m^2 c + b_{16} m^2 y + b_{17} y^2 c + b_{18} y^2 m + \\
 &\quad b_{19} c m y \\
 D_b &= c_1 c + c_2 m + c_3 y + c_4 c^2 + c_5 m^2 + c_6 y^2 + c_7 c m + c_8 c y + c_9 m y + c_{10} c^3 + \\
 &\quad c_{11} m^3 + c_{12} y^3 + c_{13} c^2 m + c_{14} c^2 y + c_{15} m^2 c + c_{16} m^2 y + c_{17} y^2 c + c_{18} y^2 m + \\
 &\quad c_{19} c m y
 \end{aligned} \tag{10}$$

where a_i , b_i , and c_i are the coefficients for each equation, and the red, green, and blue colorimetric densities are given by $D_r = \log(X_o/X)$, $D_g = \log(Y_o/Y)$, and $D_b = \log(Z_o/Z)$. The c , m and y values are the individual ink colorimetric densities. The values for these were obtained by establishing a one dimensional LUT between the dot area on film and density on paper for each primary. The a_i , b_i , and c_i coefficients in equation 10 were optimised using least-square fitting (see, for example, Lawson and Hanson²²) to give the closest colorimetric prediction to those measured. Several subgroups of samples from the cube data set were selected (e.g. using 0, 50 and 100% dot area on film of each of the three primaries in all possible combinations (27 samples required in this case as originally suggested by Clapper)) to investigate how the equation's performance is affected by the number of samples chosen for deriving it.

One of the problems with such polynomials is the possibility of local maxima and minima arising which create artifacts for local regions of colour space. If polynomials are not computed with care such effects are quite probable. Again linearising the data can play a major role here. If this is done in such a way that the magnitude of the coefficients for the higher order terms is minimised the likelihood of this giving problems is significantly reduced, though not, of course, totally eliminated. This is why basing hard copy models on density as proposed by Clapper is so useful; the addition of densities is the best simple approximation to the behaviour of changes in colorant concentration and the result of overprints made from these colorants.

3.1.3. Scanner Models

Two scanners (from Crosfield and Sharp) were selected as mentioned in section 2.1. Each device can scan both reflective and transmissive materials. Only the reflective material was investigated in this project.

a) Sharp JX600 scanner

The Kodak Q60²³ used in this study was a fore-runner to the current ANSI standard and ISO proposal for a photographic original for scanner calibration. The ANSI standard defines an image containing 166 colours with specified colorimetric values and an additional 158 colours which are material specific but for which the colorimetric values are normally provided by the supplier. The Kodak image was similar and contained 236 colours composed of 10 colours each at 12 hue angles, with variations in lightness and chroma, plus primary (cyan, magenta, and yellow), secondary (red, green and blue) and tertiary (grey) colorant scales. In addition there were twenty perceptually equally spaced neutral colours plus an extra twelve colours representing various human skin colours.

A Kodak Ektacolor Print of the Q60 was used for characterising the Sharp JX600 scanner. Each of the colours contained in it was measured using a portable spectrophotometer and telespectro-radiometer. Using the XYZ tristimulus values for each colour patch of the Q60, and resultant scanner RGB DAC values, two models were obtained based on the second and third order masking equations respectively. The RGB DAC values were normalised and transformed to CMY values, i.e. $C = 1 - R/255$. A one dimension LUT was established between the CMY values and corresponding colorimetric densities (D_r , D_g , and D_b) based on measurements made on the 20 neutral colours in the Q60 image. In this way the CMY channels were effectively linearised. Using cube root values rather than taking logarithms would have been equally effective but the latter were more convenient. All 236 colours were used to fit both the second and the third order masking equations.

b) Crosfield scanner

The Crosfield scanner was used in the mode in which it outputs the CMY, (or CMYK) DAC values directly. These values are defined empirically by an operator modifying a series of coefficients in a set of equations to produce a pleasing colour reproduction²⁴. In our studies this was done for the Cromalin proofing system. Hence, the reverse third order model (equation 10) was used as the characterisation model for the Crosfield scanner from CMY to XYZ values. (The complete transformation from CMYK to XYZ will be discussed in part two of this paper).

3.1.4 Film recorder model

To model the Fire 1000 colour transparency film recorder a range of colour patches were generated on the film recorder and their tristimulus values measured. These were converted to colorimetric density. The DAC values used to generate each patch were determined and again 'best-fit' second and third-order equations relating colorimetric density to DAC values were computed. The colour patches chosen were obtained by scanning a Q60 transparency and outputting it on the transparency recorder. The DAC values used were those obtained from the scanner and thus the characterisation of the device included the colour transformation included in the product. Using this approach provided the linearisation procedure necessary to obtain a reasonably uniformly distributed set of colours.

3.2 RESULTS

Test samples were produced on the devices to test the predictive accuracy of the above models. Six measures were employed to indicate the agreement between the models' prediction and measured

colorimetric values. These are mean values of $|\Delta x|$, $|\Delta y|$, $|\Delta u'v'|$, $\%|\Delta Y|$, ΔE (CIE $L^*a^*b^*$), ΔE (CIE $L^*u^*v^*$), and ΔE (CMC(1:1)²⁵). All are given in the various tables. However, discussion will largely be restricted to the CMC ΔE values.

In general, characterisation models were successfully derived for all devices studied in the project. The model which gave the most accurate prediction for a particular category should be recommended for use in industry. The major findings are summarised in the following sections; one for each type of device.

3.2.1 Colour Monitors

For deriving and testing performance of the models, two sets of data were produced: cube and uniform data sets including 729 and 267 colours respectively. The cube data set (see section 2.2.1) was produced using nine unequal intervals for each of the RGB channels, selected to give a reasonably uniform perceived colour difference between adjacent intervals in each channel. The uniform data set comprised 267 colours well distributed in the CIE $L^*a^*b^*$ space (with ten equally spaced metric lightness and hue intervals, plus zero to maximum displayable metric chroma). These are plotted in figure 4.

Table II summarises the predictive performance of the models developed using the cube (with unequal intervals) and the uniform data sets. The results for the PLVC model are excluded in predicting the cube data set. (This is because this data set was the LUT for this model. Hence, the predicted results would give a perfect agreement to this data set.) The results clearly indicate that all models gave very accurate predictions to the cube data set, but, conversely, very bad fit to the uniform data set, i.e. worse by at least a factor of two. In particular, the PLVC model which would be expected to give the best prediction to any data set performed the worst. Subsequent investigation showed that more than half of colours in the uniform data set were either quite dark or very saturated, i.e. the gun intensities for these were less than 0.3 (on a zero to one scale) for at least one of the three channels. This is the area of significant non-linearity in the gamma function but the visual 'spacing' of the nine unequally spaced colours for each channel had not taken account of this. (The lowest intensity was, in fact, only 0.3.) So, we had not properly linearised the device 'space'; hence the problem and a perfect example of the importance of linearisation.

New data was generated to improve the performance of these models. This included a set of 18 equally spaced colours for each gun, and a new cube data set produced using nine equal intervals. The former set was used to derive all models except the PLVC model. The cube data set was used as the LUT for the PLVC model. In addition, this LUT was determined using two types of colorimetric data: XYZ and CIE $L^*a^*b^*$. (The XYZ form was used in the earlier test).

These new versions of the models were again tested using the two original data sets. Their results are summarised in TABLE III. The results clearly show that all models gave similarly accurate predictions to the cube data set, but very large improvements for the uniform data set. The general findings are summarised below:

- The accuracy of all models was affected by the distribution of intensities chosen along each of the red, green, and blue channels. We can conclude that using either non-equal intervals, with more samples close to the low end (as suggested by Post and Calhoun¹¹), or a greater number with equal intervals, such as the 18 used in this project, should be able to derive reliable models.
- The PLVC model did not perform as well as expected. Again, because it had not been properly linearised, its accuracy was greatly dependent upon the distribution of intensities chosen for the

LUT. The model performance was slightly improved by using CIE $L^*a^*b^*$ rather than XYZ data to derive the LUT.

- The PLCC and LOG-LOG2 models gave the best overall performance, even better than that of PLVC model. They assume invariance of each gun's chromaticity coordinates. Our results suggest that phosphor constancy does hold quite well for the Barco Calibrator monitor studied. Furthermore, these models are simple to implement and should be recommended for industrial application.

3.2.2 Hard Copy Printers

For deriving and testing printer models, a cube data set was produced for each of the five devices studied. The results are summarised in Table IV, in which only the masking models derived using the subgroups including 27 and the full 729 colours are given. There are two sets of results for the Mitsubishi G650 printer. The initial set was produced using PageMaker software on a Macintosh system to drive the printer. This hard copy showed a high degree of spatial non-uniformity due to poor registration of the three primary colorants. A new printer driver was developed for a Sun SPARC workstation which enabled rotation of the printing screen angles for the three colorants. Although the spatial resolution was reduced for the output images, the colour become more uniform in comparison with that driven by the Macintosh computer.

For all five printing devices studied the third-order masking model gave the most accurate prediction to the cube data. The second-order model performed second best, followed by the modified Neugebauer, and then the original Neugebauer model (the worst). It was found that increasing the number of samples used to derive the masking model could improve its accuracy but at the cost of extra measurement. In general, there was little practical benefit gained by going beyond 64 colours, providing they were well spaced and the system reasonably well linearised. The modified Neugebauer model's performance is generally close to, though poorer than, that of the second-order masking models derived using 27 samples. This is particularly true for the low end printers, i.e. G650 and HP Deskwriter. In general, the best results were obtained for the Cromalin and IRIS systems, followed by the Canon CLC500 and the Mitsubishi G650 (new driver), the HP Deskwriter, and the Mitsubishi G650 (a Macintosh driver), the worst.

It may be noted that the low-end devices (i.e. Mitsubishi G650 and HP Deskwriter) can achieve quite good accuracy when reasonably well calibrated. The cost difference between the devices used in this project can vary by as much as five to ten times and so it is interesting to note that where usage is low such devices may prove more than adequate for digital 'proofing' purposes. However, it is important to realise that many printers have poor repeatability, particularly when different ink batches are used. Hence, it is important that they be characterised and/or calibrated regularly. if they are to be used for critical proofing applications.

3.2.3 Scanners

TABLE V gives the results for the Sharp JX600 scanner only. For the Crosfield scanner the calibration method is different due to the inclusion of a black (K) component. This will be introduced in part 2 of this paper.

Again, the results confirm the previous finding, i.e. the third-order polynomial model gave better prediction than the second-order one. The third-order model derived from the data measured using the hand-held spectrophotometer gave the most accurate prediction to the Q60 colours. Although the mean

CMC(1:1) ΔE value is 2.0, which is larger than any of the printing devices studied, the poorest predictions always occurred around the boundary of the colour gamut as expected.

The results also indicate that the models derived from the data measured using the hand-held spectrophotometer were more accurate than those obtained with the TSR. This was mainly due to the very small sample size on the Q60 image which made it very difficult, with the TSR, to set up precise measuring positions and properly take account of the illuminating/viewing geometry.

3.2.4 Film output recorder

The best model to predict the Q60 colours was again a 3rd-order polynomial. This gave a mean CIE $L^*a^*b^*$ ΔE value of around 2.6. In practice, it is known that fluctuations in processing and the output device can lead to errors of at least the order of those found in this study. Colour variations of this order can be found from batch to batch even if the device is calibrated very regularly. Hence, this performance was considered to be reasonable.

4.0 CONCLUSIONS

It is clear from these results that a full characterisation procedure, based on measurement of a large number of colours is frequently unnecessary. For colour monitors, in particular, it is only necessary to modify the conventional single exponent gamma correction to take account of the non-linearity which occurs at low luminance levels. This can be achieved quite simply by means of a look up table based on a relatively limited number of points (18 were used in this study but the number could probably be reduced at high luminance levels) or by using these points to fit a second order gamma correction. Both these approaches provided an average ΔE value, for over 1000 colours, of less than 1.0 between the measured and predicted results.

For the hard copy devices, both paper and transparency output, modelling was again found to be quite acceptable, particularly for the higher cost devices. Using second order polynomial functions (masking equations), average ΔE values of around 1.5 between predicted and measured values, for the 729 colours tested, were achieved based on measurements of less than 10% of these. Such errors are considered quite acceptable for Graphic Arts applications which are generally concerned with complex images and include processes which have a variability greater than this. The lower cost devices produced errors approximately 50% higher but even this is still quite acceptable for many applications in Graphic Arts. In general, third order polynomials reduced these errors still further (by around 30%) based on the same number of samples. The modified Neugebauer equations performed worse than the masking equations but were still reasonably acceptable and do have the advantage that they require only a limited amount of measurement data to determine. In fact the results from the modified Neugebauer equations were poorer than expected by one of the authors who had previously obtained quite acceptable results from them; better than from the masking equations.

For the transparency scanners and recorders only polynomial modelling was used in this work. Depending on the order of the polynomial, average ΔE values of 2 to 3, between predicted and measured colours, were obtained. The equations were fitted on the 236 colours of the Kodak Q60 image and the error analysis is based on the fit to these. Thus, for arbitrary colours we can expect a somewhat poorer performance. Nevertheless, for Graphic Arts applications, where the image undergoes significant gamut compression this was felt to be adequate and our practical work bore this out.

The device modelling work has demonstrated that for many purposes far fewer measurements need to be made than for full characterisation. This is of tremendous practical advantage since most devices are not stable and require fairly frequent re-calibration. This is particularly important when the addition of black ink is included as will be discussed in part 2 of this paper..

One of the dangers arising from empirically fitted polynomial functions is the possibility of local maxima and minima. However, we made extensive use of these models in our project in developing an exemplar colour management system based on this work to demonstrate its efficacy. No evidence was seen over a range of images of such artifacts. We attribute this to careful selection of the domain in which the model was fitted; thereby minimising the significance of the higher order terms. Where it may well still have introduced problems, however, was if images had been retouched in such a way that points outside the data set were defined. This was not considered in this work. Thus, we conclude that the use of polynomials is a reasonably effective and practical modelling approach but that care needs to be taken in testing their validity for more general applications.

One of the points this work has demonstrated is the importance of linearisation of the data with respect to the transformation domain. Careful attention to this can significantly reduce the amount of measurement required, even for a non-modelling approach. We attribute the success of our work here to this though we can now see how this can be improved even further. For example, the monitor PLVC model would undoubtedly improve with more careful consideration of this.

REFERENCES

1. CIE, Colorimetry, second edition, CIE Publication No. 15.2, Central Bureau of the CIE, 1986.
2. M. R. Luo, A. A. Clarke, P. A. Rhodes, A. Schappo, S. A. R. Scrivener and C. Tait, Quantifying colour appearance. Part I. LUTCHI colour appearance data, *Col. Res. & Appl.*, Vol 16, 1991, 166-180.
3. M. R. Luo, A. A. Clarke, P. A. Rhodes, A. Schappo, S. A. R. Scrivener and C. Tait, Quantifying colour appearance. Part II. Testing colour models performance using LUTCHI colour appearance data, *Col. Res. & Appl.*, Vol 16, 1991, 181-197.
4. M. R. Luo, X.W. Gao, P. A. Rhodes, J.H. Xin, A. A. Clarke, and S. A. R. Scrivener, Quantifying colour appearance. Part III. Supplementary LUTCHI colour appearance data, *Col. Res. & Appl.*, Vol 18, 1993, 98-113.
5. M. R. Luo, X.W. Gao, P. A. Rhodes, J.H. Xin, A. A. Clarke, and S. A. R. Scrivener, Quantifying colour appearance. Part IV. Transmissive media, *Col. Res. & Appl.*, Vol 18, 1993, 191-209.
6. M. R. Luo, X. W. Gao, and S. A. R. Scrivener, Quantifying colour appearance. Part V. Simultaneous Contrast, *Col. Res. & Appl.*, Vol 20, 1995, 000-000.
7. Patent
8. R.W.G. Hunt, Revised Colour Appearance Model for Related and Unrelated Colours, *Col. Res. Appl.* Vol 16, 1991, 146.

9. Hunt R.W.G., An improved predictor of colourfulness in a model of colour vision, *Col. Res. Appl.*, Vol 19, 1994, 23-26.
10. M.C. Stone, W.B. Cowan, and J.C. Beatty, Color Gamut Mapping and the Printing of Digital Color Images, *ACM Transaction Graphics*, Vol 7, 1988, 249.
11. Post D.L. and Calhoun C.S., An Evaluation of Methods for Producing Desired Colours on CRT Monitors, *Col. Res. & Appl.*, Vol 14, 1989, 172-186.
12. Cowan W.B., An inexpensive scheme for calibration of colour monitors in terms of CIE standard coordinates, *Computer Graphics*, Vol 17, 1983, 2315-321.
13. P.E. Gill and W. Murray, Quasi-Newton methods for unconstrained optimisation, *J. Int. Mat. Appl.*, Vol 9, 1982, 91-108.
14. J.W. Birkenshaw, A.J. Johnson and B.H.W. Sunderland, Colour Reproduction, *Pira Report*, 1985, Appendix 2.
15. H. E. J., Neugebauer, Die Theoretischen Grundlagen des Methirfarbendruckes, *Zeitschrift Wissenschaften Photography*, Vol 36, 1937, 73-89.
16. J. A.C. Yule, The Theory of Subtractive Colour Photography; I. The Conditions for Perfect Colour Rendering, *J. Opt. Soc. Am.*, Vol 28, 1938, 419-430.
17. A. Murray, XXXXXXXXXXXX, *J. Franklin Inst.*, Vol 221, 1936, 721.
18. J.A.C. Yule and W.J. Nielsen, The Penetration of Light into Paper and its Effect on Halftone Reproduction, *TAGA proc.*, 1951, 65.
19. J.A.C. Yule and R.S. Colt, Colorimetric Investigations in Multi-colour Printing, *TAGA Proc.*, 1951, 77.
20. F.R. Clapper, An Empirical Determination of Half-tone Colour Reproduction Requirements, *TAGA proc*, 1961, 31-41.
21. J.A.C. Yule, Principle of Colour Reproduction, John Wiley & Sons Inc., 1967, 205-232.
22. C. L. Lawson and R. Hanson, Solving Least Squares Problems, Englewood Cliffs, NJ: Prentice-Hall, 1974.
23. T.O. Maier and C.E. Rinehart, Design Criteria for an Input Colour Scanner Evaluation Test Object, *J. Photo. Sci.*, Vol 38, 1990, 169-172.
24. A.J. Johnson, Techniques for Reproducing Images in Different Media: Advantages and Disadvantages of Current Methods, *TAGA proc.*, Vol 2, 1992, 739-756.
25. F.J.J. Clarke, R. McDonald and B. Rigg, Modification to the JPC9 Colour-Difference Formula, *J. Soc. Dyers Col.*, Vol 100, 1984, 141-148.

TABLE I The imaging devices characterised in the project

CATEGORY	DEVICE
Monitor	Barco Calibrator
Printer	Cromalin, Iris Inkjet, Canon CL500, Mitsubishi G650 and HP Deskwriter
Scanner	Crosfield and Sharp JX600
Film recorder	Crosfield Magnatran (CSI 1000)

TABLE II Summary of monitor models performance (The coefficients of each model obtained using 9 unequal intervals for each gun).

Mean Measure							
Model	$ \Delta x $	$ \Delta y $	$\Delta u'v'$	$ \Delta Y $	ΔE CIE L*a*b*	ΔE CIE L*u*v*	ΔE CMC(1:1)
Cube Data Set							
PLCC	0.0010	0.0009	0.0010	0.74	0.63	0.84	0.38
LIN-LIN2	0.0025	0.0033	0.0032	1.98	1.92	2.04	1.00
LOG-LOG	0.0010	0.0012	0.0012	1.33	0.86	1.07	0.51
LOG-LOG2	0.0009	0.0009	0.0010	0.84	0.63	0.82	0.38
LOG-LIN2	0.0022	0.0030	0.0029	2.26	1.94	2.36	1.07
Uniform Data Set							
PLVC (XYZ LUT)	0.0126	0.0127	0.0137	26.13	4.93	5.21	4.45
PLCC	0.0121	0.0126	0.0133	24.30	4.76	5.06	4.25
LIN-LIN2	0.0079	0.0077	0.0080	6.07	3.40	3.76	2.37
LOG-LOG	0.0036	0.0034	0.0036	3.31	1.64	2.10	1.20
LOG-LOG2	0.0041	0.0026	0.0036	3.39	1.28	1.85	1.05
LOG-LIN2	0.0085	0.0087	0.0091	12.00	3.28	3.61	2.69

TABLE III The summary of monitor models performance. (The coefficients of each model obtained using 18 equal intervals for each gun).

Mean Measure							
Model	$ \Delta x $	$ \Delta y $	$\Delta u'v'$	$ \Delta Y $	ΔE CIE L*a*b*	ΔE CIE L*u*v*	ΔE CMC(1:1)
Cube Data Set							
PLCC	0.0011	0.0009	0.0011	0.87	0.70	0.93	0.42
LIN-LIN2	0.0036	0.0044	0.0047	4.17	2.62	2.53	1.34
LOG-LOG	0.0012	0.0013	0.0014	1.01	0.90	1.09	0.49
LOG-LOG2	0.0011	0.0010	0.0011	0.99	0.68	0.89	0.41
LOG-LIN2	0.0024	0.0029	0.0030	2.23	1.86	2.08	1.00
Uniform Data Set							
PLVC (XYZ LUT)	0.0032	0.0028	0.0034	4.24	1.28	1.51	1.09
PLVC (LAB LUT)	0.0031	0.0027	0.0033	2.69	1.19	1.43	0.88
PLCC	0.0020	0.0025	0.0026	2.81	0.99	1.30	0.83
LIN -LIN2	0.0071	0.0064	0.0072	6.05	2.60	2.77	2.05
LOG -LOG	0.0026	0.0038	0.0035	3.34	1.52	1.81	1.12
LOG - LOG2	0.0026	0.0025	0.0029	3.57	1.02	1.34	0.90

TABLE IV Summary of printing models performance for each printing device studied.

Model	□ □ □	□Δx	Δy	Δu'v'	% ΔY	ΔE CIE L*a*b*	ΔE CIE L*u*v*	ΔE CMC(1:1)
Cromalin Proofing system								
Masking Equations								
Second-order	27	0.0022	0.0030	0.0029	2.02	2.38	1.87	1.33
Second-order	729	0.0021	0.0024	0.0025	1.81	2.01	1.56	1.07
Third-order	27	0.0017	0.0020	0.0019	1.57	1.48	1.18	0.81
Third-order	729	0.0014	0.0016	0.0015	1.35	1.14	0.91	0.63
Neugebauer equations								
Unmodified	8	0.0095	0.0058	0.0070	5.37	5.72	4.27	3.06
Modified	8	0.0035	0.0042	0.0038	4.17	3.43	2.75	1.90
IRIS Ink Jet								
Masking Equations								
Second-order	27	0.0026	0.0042	0.0036	2.40	3.25	2.33	1.61
Second-order	729	0.0021	0.0027	0.0026	1.69	2.27	1.61	1.08
Third-order	27	0.0023	0.0019	0.0022	2.23	2.05	1.47	1.00
Third-order	729	0.0016	0.0012	0.0015	0.95	1.36	0.92	0.62
Neugebauer Equations								
Unmodified	8	0.0131	0.0098	0.0117	2.95	10.90	7.03	4.53
Modified	8	0.0047	0.0061	0.0051	5.88	5.01	3.75	2.38
CANON CLC500								
Masking Equations								
Second-order	27	0.0030	0.0035	0.0038	3.56	3.20	2.42	1.62
Second-order	729	0.0027	0.0032	0.0035	3.02	2.79	2.15	1.45
Third-order	27	0.0027	0.0033	0.0034	3.27	2.94	2.26	1.50
Third-order	729	0.0020	0.0025	0.0026	2.32	2.04	1.59	1.07
Neugebauer Equations								
Unmodified	8	0.0109	0.0054	0.0092	5.45	7.77	5.32	3.66
Modified	8	0.0042	0.0047	0.0049	5.65	4.71	3.59	2.40
G650 (a Macintosh driver)								
Masking Equations								
Second-order	27	0.0071	0.0065	0.0069	4.54	5.58	3.92	2.82
Second-order	729	0.0056	0.0062	0.0057	4.08	4.67	3.40	2.51
Third-order	27	0.0055	0.0062	0.0056	4.11	4.46	3.23	2.25
Third-order	729	0.0037	0.0040	0.0040	3.52	3.26	2.41	1.75
Neugebauer Equations								
Unmodified	8	0.0074	0.0080	0.0079	5.00	6.42	4.72	3.28
Modified	8	0.0057	0.0069	0.0060	5.35	5.02	3.78	2.68
G650 (a new driver)								
Masking Equations								
Second-order	27	0.0042	0.0052	0.0051	3.71	4.30	3.23	2.24
Second-order	729	0.0032	0.0036	0.0037	3.27	2.98	2.30	1.61
Third-order	27	0.0030	0.0026	0.0031	3.62	2.60	1.97	1.48
Third-order	729	0.0022	0.0019	0.0022	2.73	1.92	1.45	1.08
Neugebauer Equations								
Unmodified	8	0.0080	0.0047	0.0073	7.45	5.40	4.24	3.00
Modified	8	0.0058	0.0052	0.0058	6.90	4.34	3.63	2.51
HP DESKWRITER								
Masking Equations								
Second-order	27	0.0059	0.0072	0.0073	5.09	5.95	4.57	2.87
Second-order	729	0.0043	0.0049	0.0050	4.16	3.89	3.07	1.92
Third-order	27	0.0047	0.0043	0.0048	3.21	4.23	3.20	1.91
Third-order	729	0.0027	0.0027	0.0030	2.49	2.56	1.94	1.23
Neugebauer Equations								
Unmodified	8	0.0146	0.0124	0.0146	6.39	12.71	9.53	5.67
Modified	8	0.0057	0.0088	0.0080	6.81	6.75	5.64	3.12

TABLE V Summary of Sharp JX600 scanner performance using 236 colours on the Q60 image.

Model	$ \Delta x $	$ \Delta y $	$\Delta u'v'$	$\% \Delta Y $	ΔE CIE L*a*b*	ΔE CIE L*u*v*	ΔE CMC(1:1)
Portable spectrophotometer							
Second-order	0.0045	0.0060	0.0057	3.84	3.35	2.84	2.28
Third-order	0.0038	0.0050	0.0046	3.19	2.68	2.32	1.96
Telespectroradiometer							
Second-order	0.0050	0.0081	0.0063	2.80	4.79	3.68	3.04
Third-order	0.0041	0.0069	0.0052	2.63	3.85	3.00	2.56

ASPECTS OF COLOUR MANAGEMENT

Part 2 - CHARACTERISATION OF FOUR COLOUR IMAGING DEVICES AND COLOUR GAMUT COMPRESSION

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1.0 INTRODUCTION

In part 1 of this paper we discussed the characterisation of three colour imaging devices, colour scanners, colour monitors and hard copy printers, including transparency recorders. However, the majority of Graphic Arts printing devices also use a black ink in addition. There are four main advantages in doing this¹. These are:

- i) To make the control of the neutral balance between the three colours less critical,
- ii) To produce darker shadows with better detail,
- iii) To replace the more expensive process colours with the relatively cheap black,
- iv) To avoid the piling up of several inks during printing, thereby reducing ink drying problems in a high-speed printing process.

However, black ink degrades the reproduction quality when the tone reproduction of the black is not well controlled for the particular process for which it is being made. Therefore, it is necessary to define the optimum tone reproduction curve of the black separation for achieving high colour fidelity reproduction. Unlike a three-colour process, where there is only one combination of the three inks which will produce a given colour, a four-colour process can produce all except the cleanest colours either with a minimum amount of the three colours and a maximum amount of black or vice versa. Clearly, there are many intermediate combinations between these two extremes. This results in many different potential choices for the black ink calculation and so some rules need to be determined. To a large extent these depend upon the printing process; for some processes it is desirable to only add sufficient black to make the dark colours darker whereas for others it is useful to use black ink as a replacement for the grey component produced by only using the three chromatic inks. This is known as grey component replacement (GCR).

In section 2 of this part of the paper we extend the three colour device characterisation procedure discussed in part 1 to the more typical four colour devices. It is convenient when doing this to treat the calculation as a serial process. Initially, it is considered as trichromatic since it then leads to a unique solution for every reproducible colour. The black ink required to extend the colour gamut is then calculated followed by the level necessary for the replacement of the three coloured inks by black where appropriate. Part 1 of this paper dealt with the calculation of the unique trichromatic match whilst this part covers the black ink generation. It should be noted that the paper does not address the problem of defining which of the infinite number of solutions to the problem should be selected. What is done is to investigate models which describe the behaviour of adding black ink to chromatic inks. In order to define a specific type of black printer (i.e. gamut extension only, or combined with some specific level of GCR) it is necessary to introduce additional rules. These are beyond the scope of this work, though they are a fairly obvious extension of it. In our demonstrations of these studies, since we calculated the trichromatic match first and then computed the effect of adding black, what was achieved was gamut extension. To provide GCR it would simply be necessary to compute the grey component of a colour and use the first of the models described to decide how to replace it by black. This, and simple extensions to it have been discussed by Johnson².

The four stage transform described in part 1 of this paper presupposes that all the colours which have been generated can be rendered. This is often not the case and colour gamut compression is then required. This problem has been addressed in this project and it essentially defines a new stage in the transform in which the hue, lightness and chroma values are compressed into those which can be rendered prior to calculation of the rendering tristimulus values. This is discussed in section three.

2. BLACK PRINTER ALGORITHMS

When three or four inks are superimposed, it is sometimes assumed that the resultant density would equal the sum of the densities of the individual inks. This is known as the additivity rule. However, in practice, the overall density is usually much less than the sum of the these inks. This deficiency, called additivity failure, is discussed extensively by Yule¹. It is caused by several factors including first-surface reflection, multiple internal reflection, opacity of the inks, trapping and back transfer effects of the inks, sideways scattering of light in the paper, the halftone structure of the printed dots, the spectral absorption characteristics of the ink and paper and spectral sensitivity of the measuring instruments. These deficiencies can be characterised by mathematical models. Four of these were developed for this study as discussed below.

Two sets of calibration data were produced and measured for deriving black printer algorithms. These are denoted as the 110 and 31 sample sets. For the 110 sample set, one hundred colours were produced with ten levels of approximately equal amounts of the coloured inks each combined with ten levels of black. The ten levels of black ink were also printed on their own. For the 31 sample set, all colours included black ink together with two of the CMY inks. The former set of colours were all approximately neutral; the latter were as colourful as it is possible to be with a black ink included.

Two printing devices were investigated, Cromalin proofing and IRIS inkjet. As in part 1 of this paper, only the forward models (predicting XYZ tristimulus values from a given set of CMYK primaries) are described here. As stated earlier, the reverse process needs additional information to define the particular solution required. This process is well understood and therefore not investigated in our work.

2.1 SIMPLE ADDITIVITY FAILURE MODEL (SAFM)

The colorimetric densities (D_r , D_g and D_b) for each sample in the 110 sample set were computed. As shown in figure 1 regression lines were obtained by plotting the densities from the one hundred colours produced using three coloured inks plus black against those of the black ink alone. For each colour channel ten regression lines were plotted; one for each level of three-colour component with variations in black content. If there were no additivity failure all the regression lines would be parallel; however, for halftone printing it is common to find that the lines approximately converge to a point on the 45° straight line passing through the origin as shown in figure 2. This convergence density is 2.2 for the data of figure 1. The mathematical expressions defining this behaviour are given in equation 11.

$$\begin{aligned} D_{r-4c} &= D_{r-3c} + D_{r-k} - (D_{r-3c} D_{r-k}) / k_r \\ D_{g-4c} &= D_{g-3c} + D_{g-k} - (D_{g-3c} D_{g-k}) / k_g \\ D_{b-4c} &= D_{b-3c} + D_{b-k} - (D_{b-3c} D_{b-k}) / k_b \end{aligned} \quad (11)$$

where D_{4c} , D_{3c} and D_k values are the colorimetric densities of the four-colour, three-colour, and black components respectively and k is the colorimetric density of the convergence point as described above.

2.2 MODIFIED ADDITIVITY FAILURE MODEL (MAFM)

The results obtained from the SAFM were further modified to derive a factor k' to replace k in equation 11. This factor was defined by the following equation:

$$k'_r = a_{1,1} D_{r-3c} + a_{1,2} D_{r-k} + a_{1,3} D_{r-3c}^2 + a_{1,4} D_{r-k}^2 + a_{1,5} D_{r-3c} D_{r-k} + a_{1,6} D_{r-3c}^3 + a_{1,7} D_{r-k}^3 + a_{1,8} D_{r-3c}^2 D_{r-k} + a_{1,9} D_{r-k}^2 D_{r-3c} + a_{1,10} \quad (12)$$

where $a_{i,j}$ are the optimised coefficients ($i = 1$ to 3 for D_r , D_g and D_b respectively), obtained by the method of least squares. D_{r-3c} , D_{r-k} , D_{r-4c} are the red-colorimetric densities for the three-colour, black and four-colour components respectively. Similar equations are necessary for k'_g and k'_b .

2.3 MASKING EQUATIONS

Rather than model additivity failure directly an alternative approach is to empirically fit a polynomial function which may correct for other sources of error also, such as proportionality failure. Such an approach has already been considered for the three colour case and can be extended to four colours as proposed by Clapper³. We did not follow his approach exactly but evaluated two alternative polynomial functions as described in the following sections.

2.3.1 Third-order Polynomial Model (3PM)

A simple third-order polynomial was derived as shown in equation 13, with no cross products between channels.

$$D_{r-4c} = a_{1,1} D_{r-3c} + a_{1,2} D_{r-k} + a_{1,3} D_{r-3c}^2 + a_{1,4} D_{r-k}^2 + a_{1,5} D_{r-3c} D_{r-k} + a_{1,6} D_{r-3c}^3 + a_{1,7} D_{r-k}^3 + a_{1,8} D_{r-3c}^2 D_{r-k} + a_{1,9} D_{r-k}^2 D_{r-3c} + a_{1,10} \quad (13)$$

where a_{ij} are the optimised coefficients ($i = 1$ to 3 for D_r , D_g and D_b respectively). D_{r-3c} , D_{r-k} , D_{r-4c} are the colorimetric densities for the three-colour, black-ink and four-colour components respectively. Similar equations are necessary for D_{g-4c} and D_{b-4c} .

2.3.2 Second-order Polynomial Model (2PM)

This model is similar to the 3rd-order polynomial model except that the equations include cross product terms of D_r , D_g , and D_b but no third order terms. The forward model is given in equation 14 with the same definitions of the parameters and coefficients as equation 13.

$$D_{r-4c} = a_{1,1} D_{r-3c} + a_{1,2} D_{g-3c} + a_{1,3} D_{b-3c} + a_{1,4} D_{r-k} + a_{1,5} D_{g-k} + a_{1,6} D_{b-k} + a_{1,7} D_{r-3c}^2 + a_{1,8} D_{g-3c}^2 + a_{1,9} D_{b-3c}^2 + a_{1,10} D_{r-k}^2 + a_{1,11} D_{g-k}^2 + a_{1,12} D_{b-k}^2 + a_{1,13} D_{r-3c} D_{g-3c} + a_{1,14} D_{r-3c} D_{b-3c} + a_{1,15} D_{g-3c} D_{b-3c} + a_{1,16} D_{r-3c} D_{r-k} + a_{1,17} D_{r-3c} D_{g-k} + a_{1,18} D_{r-3c} D_{b-k} + a_{1,19} D_{g-3c} D_{r-k} + a_{1,20} D_{g-3c} D_{g-k} + a_{1,21} D_{g-3c} D_{b-k} + a_{1,22} D_{b-3c} D_{r-k} + a_{1,23} D_{b-3c} D_{g-k} + a_{1,24} D_{b-3c} D_{b-k} + a_{1,25} D_{r-k} D_{g-k} + a_{1,26} D_{r-k} D_{b-k} + a_{1,27} D_{g-k} D_{b-k} \quad (14)$$

2.4 RESULTS

The first model was defined from the 110 neutral samples only. The last three models were defined from all 141 colours including Black. The models were tested using the 31 and 110 black calibration sets for both the Cromalin and IRIS devices. The results are summarised in TABLE I in terms of mean CMC(1:1) ΔE computed between the models' predicted and measured results. The results indicate that when comparing the various forward models, the 2PM performed the best and the SAFM the worst. The 2PM was also used to characterise the Crosfield scanner as mentioned in section 3.2.3.

3. GAMUT COMPRESSION

During the process of colour reproduction it is rare that any device and/or media on which an image is being reproduced can reproduce all the colours of the original. In some instances the constraints of the device are relatively small; transparency film materials, for example, can reproduce most of the colours which occur naturally, at least in terms of relative measures such as lightness and chroma.

In other situations the device colour gamut limitations are quite severe. Newspaper reproduction, for example, provides a very limited gamut of colours and when any photographic image is reproduced in

newspapers a significant compression of the gamut is required. In general, any printed reproduction of a photographic or colour monitor original requires some colours to be rendered which are not achievable on the hard copy device. Within much of the Graphic Arts industry such compression is undertaken empirically using algorithms which allow the user to modify local regions of colour space as discussed by Johnson². However, this is usually achieved in the device colour space and the question arises as to how this is best achieved within a colorimetric domain.

In part 1 of this paper the concept of the four-stage transform was introduced. This consists of a conversion from device space to tristimulus values, followed by transformation to an appearance space, both based on the viewing conditions for the original image. These perceptual attributes are then converted back to tristimulus values but this time taking account of the viewing conditions for the reproduction and finally transformed into the output device drive values.

To take account of gamut compression this model needs to be extended and the logical place to do this is within the calculation of the perceptual attributes. Effectively we need a function which transforms the perceptual attributes of the original image into a different set which provides optimum gamut compression prior to calculation of the output tristimulus values and drive values. In order to find a working hypothesis which could be evaluated we reviewed the available literature, as discussed below.

In the Graphic Arts industry the most extensive work on preferred reproductions was started by Yule⁴ and continued later with his co-workers Pobboravsky and Pearson⁵⁻⁹ between 1968 and 1975. Johnson¹⁰ reviewed their work and chose to ignore the apparent anomalies such as hue shifts and lightness/saturation compression being non-uniform speculating that they may be artifacts of their experimental method. On the basis of the remaining work, together with that of other publications, he proposed that gamuts should be compressed according to the following rules:

- a) Maintain hue - there seemed no logical reason, at that time, for changing hue of a colour.
- b) Compress perceived lightness linearly - this was based on earlier findings¹¹ which suggested that optimum reproduction is achieved when lightness is computed via the Bartleson-Breneman equation, for the viewing conditions appropriate for that original, and linearly compressed into print lightness.
- c) Compress perceived colourfulness (chroma) linearly - this was an unproven hypothesis based on the limited information available from the literature. At that time the best formula available for perceived colourfulness was that proposed by Hunt¹².

A small number of images were made to these criteria and appeared to reproduce reasonably well. However, since that time further studies, combined with considerable practical experience, have shown some limitations in this approach. Laihanen¹³ claimed to have had difficulties with such an approach in that it worked well for some images but not others. He stated that this is due to the different shapes of the colour solids and suggested he obtained better results by compressing the data in such a way that the correction vector was made to map toward the centre of the lightness scale. However, the difficulty in understanding his approach is that he shows no lightness compression requirement which is somewhat surprising. We can only assume that the lightness scales were normalised prior to compressing the chroma. This seems a quite reasonable approach but fails to explain an important part of the compression algorithm without discussion as to how the normalisation was done.

Viggiano and Wang¹⁴ undertook a study which ended up concluding that the optimum compression was obtained by the original hypothesis above; namely linear compression of lightness and chroma whilst maintaining hue. However, since they obtained their samples on monitors it seems likely that the 'shapes' of the gamuts were similar in all cases. As already stated, Laihanen found that this procedure

failed most noticeably when the gamut 'shapes' were different. Stone et al¹⁴, addressing the problem from the point of view of computer graphics where the image on the monitor is the original, suggested scaling the neutral axis such that black and white on the original were modified to map into the black and white of the reproduction and then modifying the chromaticity co-ordinates of the monitor RGB values such that they came within the gamut of the output device.

It is rather disappointing that despite many years of colour reproduction practice, there are few published results to show how gamut compression should be quantified. What is interesting to note from the studies above is what does not work, and what reasons there may be for this. We decided to investigate the limited proposals discussed above by an experimental study of what is current practice in graphic arts. The experiment undertaken for this work was to take a Q60 and have it scanned on a number of scanners by different operators. Since scanners from three different manufacturers were used, each operated by different individuals, we could be certain that we were not providing a bias introduced by a particular scanner manufacturer or operator. Furthermore, since we asked that the company doing the scanning also made the proofs we could be certain that the operators were working in the environment in which they felt most familiar. By using experienced operators, who earn their living by achieving good gamut compression day in and day out, we believe that this provided the best possible test of the hypotheses found.

3.1 EXPERIMENTAL DETAILS

A Kodak Q60 Ektachrome transparency (as described in part 1 of this paper) was taken to five different sites for reproduction. Two of these sites were in the UK and three in the USA. Across the five sites, the results were obtained from two Crosfield scanners, two from Hell and one from Scitex. Each operator was asked to scan the job twice. The first time he should use his "standard" setup. The second time he should edit this setup if he felt this would provide improvements to the result. Thus ten scans of the Q60 and proofs were made for each site.

The proofs were measured, using a hand-held spectrophotometer with 45/0 measurement geometry and CIE L^* , a^* , b^* , C_{ab}^* and h_{ab} were computed. The transparency was measured using a telespectroradiometer and the same parameters computed. Since the sample was placed on a diffusing viewer the transmission measurement geometry was diffuse/specular. All values were initially computed using the perfect diffuser/transmitter under illuminant D50 as the reference white for the 2° observer.

3.2 RESULTS

For each of the proofs L^* , C_{ab}^* and h_{ab} respectively were plotted as functions of the same parameters measured on the transparency, for each of the 224 colours in the image. One of the plots is shown in figure 3. Thus, if all the colours in the proof and transparency had identical tristimulus values the plots would have produced 45° lines. In addition, for one of the data sets, the same plots were produced but each concentrating on specific regions of colour space in order to try to understand some anomalous behaviour, when compared to the hypothesis discussed earlier.

For all the proofs data was plotted relative to the perfect transmitter/diffuser. However, this is not the normal manner for evaluating and reproducing images. There is often considerable debate amongst those studying image reproduction over what the reference should be but the experience of these authors suggests that the most useful comparisons are often achieved when the reference white is selected from within the image, when it exists. The results for the same set of data were therefore modified to make the lightest step of the grey scale 100 for both print and proof.

In addition, for the pure primary and secondary colorant scales in the Q60, lightness was plotted as a function of chroma for the same single set of data but in both absolute and relative modes. An example of this is given in figure 4.

Also, for the same primary and secondary colorant scales as well as for columns 1, 3, 4 and 5 of the Q60, b^* was plotted as a function of a^* . Since the normalised lightness v chroma and $a^* v b^*$ plots for the primary and secondary colours were particularly informative these were plotted for all of the remaining nine data sets. Again, an example of the results obtained is given in figure 5.

3.3 DISCUSSION OF RESULTS

All the complete sets of results showed a number of similar trends although there were some differences between each. For lightness, as defined by L^* , there were two significant points. There were a range of colours, generally of fairly high lightness, which were reduced in lightness on the proof over the general increase in lightness which was observed. For some of the proofs the effect was greater than the others but it was quite distinct. Furthermore, if we ignored these colours, although L^* provided a reasonably linear relationship it was clear that a nonlinear modification to L^* , such as the Bartleson-Breneman equation, would improve the correlation.

Chroma had a significant degree of scatter on all sets. There was no clear evidence of any uniform compression of chroma across all hues.

There were clearly hue shifts which were common to all the results. Cyan and green colours were generally reproduced somewhat bluer and blue colours somewhat greener. Other than this the reproduced hue was generally reasonably accurate on the proofs made on the Hell and Crosfield scanners although there were small individual differences which were not common. For example, one operator did not seem to like the yellow greens being quite as green as many of the others and another had made some hue shifts toward blue in the magenta region as well. The Scitex scans showed some rather more significant deviations in hue for the magenta and red colours.

The "white point normalised" results showed the same sort of pattern although there was a distinct reduction in chroma for the proofs when compared to the transparency. However, the degree of scatter was as described above.

For the plots of a^* versus b^* for the primary and secondary colours a number of hue shifts were evident. The red and yellow scales were reproduced quite well but the cyan and, to a lesser extent, the magenta showed significant shifts. The green and blue were reasonably good for low chroma values but the error increased with chroma. The same comments applied to the relative values also.

The a^* vs. b^* plots of columns 1, 3, 4 and 5 of the Q60 showed the effect as the chroma of the colours on the transparency was reduced. For the high chroma colours, which were closer to the transparency gamut boundary, the change required was far greater than for the low chroma values.

The lightness versus chroma plots showed a surprising result. In general the proof gamut lay outside of the transparency gamut. This is totally contradictory to experience and indicated that the definition of chroma and/or lightness was inadequate. However, when looking at the relative values this situation was reversed. Thus this adds credence to the argument that for most purposes it is more useful to make assessments on relative rather than absolute measures.

In reviewing the relative data it was clear that the yellow scale was only compressed in chroma. This was not too surprising since there was little scope for a lightness change. The blue, green and magenta on the other hand showed a compression in both chroma and lightness the direction of which, particularly for the latter two colours, could be approximated by constructing a line between the transparency gamut and the neutral grey with 50% lightness. The cyan and red exhibited a compression that was dominated by a change in lightness although the latter was only relatively small since the gamuts were quite similar except for the high chroma colours.

Another point to note was that the value of highest chroma on the transparency was invariably mapped to the value of highest chroma on the print. This happened whether the transparency gamut enclosed the proof gamut or whether the proof gamut was partly outside. It was also interesting to note that the differences in chroma between the steps of the scale seemed to be approximately linearly scaled for each colour and the nonlinear change was primarily in lightness.

These results suggested a mapping algorithm for mapping one gamut to another which is similar to that used on scanners in that it treats the primary and secondary hues separately and combines these for intermediate hues. The algorithm could be simplified if the hue were maintained but this is probably not reasonable for most colour domains. The details of the compression almost certainly require prior knowledge of the image source since this will define the precise nature of the compression construction although it is likely that a relatively small number can be defined since the image sources and reproduction media (for 4 colour printing) are also few in number. The complexity of the algorithm can be increased to suit any particular process but the following suggests itself as a reasonable approach for compressing Ektachrome colours to the printing gamut.

- a) Determine lightness compression on a grey scale which maps white to white and black to black. Scale linearly using L^* or, preferably, Bartleson-Breneman lightness.
- b) Establish the chroma and any additional lightness compression for chromatic colours by determining the gamut boundaries at each of the six hues represented by primary and secondary colours and use the following algorithm:
define cusp of each plot.
 - if one gamut totally encloses another, and the lightnesses of the cusps are similar, define the lightness value of the cusp of the reproduction gamut and "construct" radial lines from the point which has the same lightness but where chroma is zero. This will define the direction of the correction for these hues. If the lightnesses of the cusps are not similar map one to the other and extrapolate this line to the lightness axis. Construct radial lines from the resulting point.
 - if one gamut does not enclose another (as for cyan and red in this experiment) define the direction of correction by a series of radial lines which map to 50% of maximum chroma at that hue.
- c) The amount of correction along any line is given by $x(c-b)/(c-a)$ where x is the distance along the radial line, c is the point on the outer gamut and b on the inner gamut along this same line and a is the point of intersection with the neutral axis. This is a simple expression that has the effect of reducing the correction as chroma decreases. There may be some merit in making this more nonlinear; we had insufficient data to be precise about this since the Q60 only has 3 levels of chroma for each of three lightness levels at each hue angle.
- d) For each of the six primary and secondary colours determine the hue shift between them. Use this to 'correct' the hue for each colour by translating original hues to those of the reproduction.

Our original conclusion was that the correction should be a complete hue correction but subsequent study suggests this may be excessive when the difference is large. In such cases a partial correction (say, 0.5) may be preferable.

- e) For intermediate hues the hue, chroma and lightness shift will be the proportion of the correction of the two of the six hues which lie either side of the sample. (Clearly, such a procedure will only work if the boundaries are 'well behaved', e.g. not severely 'concave'. If this is not the case the number or choice of hues may need to be modified).

4 CONCLUSIONS

Testing of the four black printer algorithms demonstrated that the simple model of additivity failure which shows a set of lines converging to a point is reasonably effective but can be improved. We also know from other studies that this behaviour is less well followed when the ink density is high. Other techniques were investigated in order to improve the prediction based on the higher order masking equations. A simple third order model without cross-products was evaluated as well as a second order model with all the cross-product terms. The latter gave us the best prediction. This indicates that the departures from linearity are fairly complex. The fit obtained was considered reasonable from all models for the more neutral colours, but for the more chromatic ones the improvement from the second order polynomial was significant.

In considering the reverse model the problem arises that the solution is not unique. This means that additional constraints are required. In this work we assumed no GCR which meant that the black was only added when the gamut limit was reached with the three colour solution. Although such a solution was adequate for our purposes it would not meet normal printing requirements. However, the extension of our work to achieve this would not be difficult. The exact nature of the solution can be complex if a particular GCR requirement is defined for each lightness level but normally it does not need to be that specific. Providing the amount of ink used is reduced at the darkest tones to the level needed for satisfactory ink drying the rest of the tonal range is less critical. Thus, a solution in which the grey component for the whole scale is reduced in the same proportion as that for the shadows is adequate, whatever domain is used. Taking the polynomial approach alone would require a rather complex iterative procedure to define the correct ratio of black to coloured inks for a specific printing condition. However, use of the simple additivity failure model permits determination of this ratio more simply. Once the ratio is known it is a simple matter to define the proportion of the grey component to be subtract from the grey and apply this ratio to the whole image. A unique solution is then defined.

With regard to gamut compression a literature review had indicated that there has been no definitive colour gamut compression algorithm published which appears to be generally satisfactory. Under certain conditions a simple set of rules seemed to provide satisfactory results, namely:

- Compress perceived lightness linearly
- Maintain hue
- Compress perceived chroma linearly

However, when the shape of the original gamut is not similar to that of the reproduction this set of rules does not provide the desired result. In particular, the requirement to compress chroma differently in different regions of colour space needs to be achieved, the need to maintain hue must be relaxed and modifying the lightness of the chromatic colours differently to that of the neutrals is required.

We verified from a series of scans of the same original that gamut compression in the graphic arts industry follows a generally similar set of rules, regardless of the type of equipment used for colour reproduction or the operator doing the work. The results from this study suggest that to construct the algorithm for any material/device pair it is necessary to define the white and black points of each and compute the gamut boundaries of each of the six hues specified. From this the preferred compression can be computed in such a way that it provides results similar to those found in our experiments. This involves correction of hues towards those of the reproduction medium; mapping original white and black to reproduction white and black respectively; non-linear compression of lightness within these limits to take account of the viewing conditions for the original media and compression of chroma based on the mapping of maximum chroma of the original to that of the reproduction, for each of the primary and secondary hues.

REFERENCES

1. J.A.C. Yule, Principle of Colour Reproduction, John Wiley & Sons Inc., 1967, 205-232.
2. A.J. Johnson, Techniques for Reproducing Images in Different Media: Advantages and Disadvantages of Current Methods, TAGA proc., Vol 2, 1992, 739-756.
3. F.R. Clapper, An Empirical Determination of Half-tone Colour Reproduction Requirements, TAGA proc, 1961, 31-41.
4. Yule, J.A.C. , Proc. of the Seventh International Conference of Printing Research Institutes, Pergamon Press.
5. Pobboravsky, I., Pearson, M. and Yule, J.A.C., Computation of the halftone colour gamuts of process inks, TAGA Conf. Proc., p331
6. Pobboravsky, I., Pearson, M. and Yule, J.A.C., The colour gamut of process inks computed by the Neugebauer equations, Print. Tech. 12, 3, p150
7. Pobboravsky, I., Pearson, M. and Yule, J.A.C., The relationship between photomechanical colour reproductions and the original copy, ISCC. Conf. Proc., p159
8. Pobboravsky, I., Pearson, M. and Yule, J.A.C., Comparison of originals and printed colour reproductions, Eleventh IARIGAI Conf. Proc., p73
9. Pobboravsky, I., Pearson, M. and Yule, J.A.C. Design of a printed spectrum using modified Neugebauer equations, J. Opt. Soc. Am., 65, 3, p323
10. Johnson, A.J., Perceptual requirements of Digital Picture Processing, Paper presented at IARIGAI symposium and printed in part in Printing World Feb. 6th 1980
11. Bartleson, C.J. and Breneman, E.J., Brightness perception in complex fields, J. Opt. Soc. Am., 57, 7, p953
12. Hunt, R.W.G., Specifying colour appearance, Light. Res. and Tech., 11, 4, p175
13. Laihanen, P., Colour reproduction theory based on the principles of colour science, IARIGAI Conf. Proc., 19, p1

14. Viggiano, J.A.S. and Wang J., A comparison of algorithms for matching colour between media of different luminance ranges, TAGA Conf. Proc. (Vol. 2), p959
15. M.C. Stone, W.B. Cowan, and J.C. Beatty, Color Gamut Mapping and the Printing of Digital Color Images, *ACM Transaction Graphics*, Vol 7, 1988, 249.

TABLE I Summary of black ink algorithms performance in terms of the CMC(1:1) ΔE values.

Model	SAM	MAFM	3PM	2PM
110 Sample Set				
IRIS	2.48	1.98	1.70	1.72
Cromalin	1.31	2.02	1.89	1.23
31 Sample Set				
IRIS	3.13	2.55	2.47	1.72
Cromalin	3.51	3.07	3.26	2.73